

# DataScience Lab

## Robust classification

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① DATA AUGMENTATION

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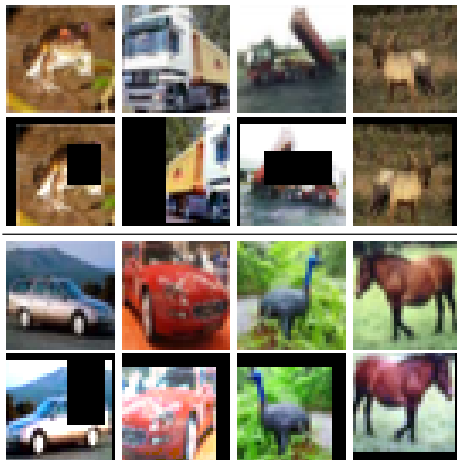
④ PERSPECTIVE AND CONCLUSION

# DATA AUGMENTATION

## Data Augmentation

### Applied Transformations :

- Random Crop
- Random Horizontal Flip
- Slight Color Jitter
- RandAugment Policy from Cubuk-2019 : [1]
- Random Erasing



## STRUCTURE AND TRAINING

## Model Architecture : Standard 3-Block CNN

- **3 CNN Blocks**
- Each block followed by : **Batch Normalization** and **Dropout**.
- **Classification Head** : A **Fully Connected** classifier block with **Dropout**.
- For Wavelet model we replace the 3 CNN blocks by the wavelet feature extractor

# Adversarial training

## PGD loss

$$\mathcal{L}_{\text{adv}}(\theta) = \mathbb{E}_{(x,y)} [\mathcal{L}(f_{\theta}(x^{\text{adv}}), y)] .$$

## TRADES loss [7]

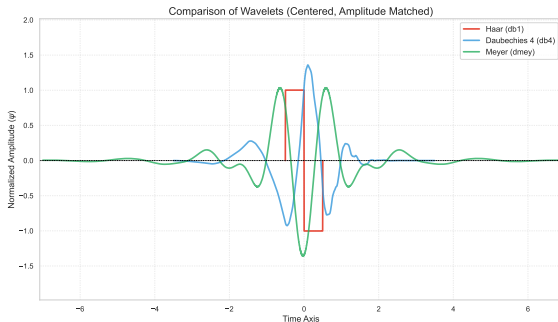
$$\mathcal{L}_{\text{TRADES}}(\theta) = \mathbb{E}_{(x,y)} \left[ \underbrace{\mathcal{L}(f_{\theta}(x), y)}_{\text{natural accuracy}} + \beta \cdot \underbrace{\text{KL}(f_{\theta}(x) \parallel f_{\theta}(x^{\text{adv}}))}_{\text{robustness penalty}} \right]$$

## Feature extraction with Wavelets [4]

### Brief definition

- $\psi \in L^2(\mathbb{R})$  is **localized** and **oscillatory** (band-pass filter), satisfying :  $\int_{\mathbb{R}} \psi(x) dx = 0$ .
- Wavelet Basis  $\Psi = \{\psi_{j,k}\}_{j,k \in \mathbb{Z}^2}$  is generated through **dilation** ( $j$  : scale) and **translation** ( $k$  : position) :

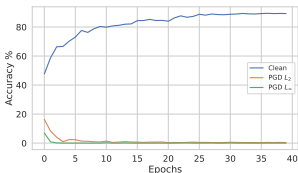
$$\psi_{j,k}(x) = 2^{-j/2} \psi(2^{-j}x - k).$$



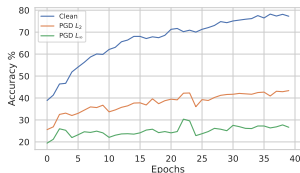
## Theoretical Advantages of Wavelets for Feature Extraction

- **Intrinsic Stability to Perturbations** : The Wavelet Scattering Transform (WST) is **provably Lipschitz continuous** w.r.t. geometric distortions (diffeomorphisms) [5].
- **Optimal Sparse Representation** : Wavelet coefficients characterize **Besov regularity** ( $B_{p,q}^s$ ), leading to efficient, **entropy-optimal** feature extraction [4].
- **High-Frequency Noise Filtering** : Multiscale decomposition naturally isolates and suppresses non-semantic noise (e.g., adversarial perturbations of high frequency), enhancing robustness by design [6].

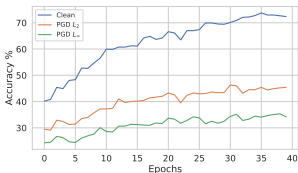
# Experiments



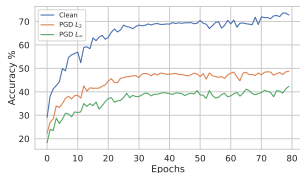
No adversarial training



FGSM  $L_\infty$  Adversarial Training



10-iteration  $L_\infty$ -PGD with CE loss



10-iteration  $L_\infty$ -PGD with TRADES loss

# Performances

| Method                   | Acc Nat   | Acc PGD $\ell^2$ | Acc PGD $\ell^\infty$ | Aggregated |
|--------------------------|-----------|------------------|-----------------------|------------|
| FGSM $\ell^2$            | <b>81</b> | 32               | 12.6                  | 44.6       |
| FGSM $\ell^\infty$       | 78        | 42               | 25.7                  | 67.7       |
| PGD $\ell^\infty$ CE     | 72        | 45               | 34                    | 79.0       |
| PGD $\ell^\infty$ TRADES | 73        | <b>49</b>        | <b>42</b>             | <b>101</b> |
| Wavelet                  | *         | *                | *                     | *          |

## PERSPECTIVE AND CONCLUSION

## Perspective and conclusion

- **Data augmentation** Explicit adversarial defense is indispensable. Strong data augmentation alone does **not** suffice.
- **TRADES** The TRADES objective is the most effective among the ones tested : more stable and robust than standard PGD-based adversarial training.
- **Future perspective : Wavelet feature extraction** Using Meyer's wavelet (to ensure differentiability) remains the most promising direction for achieving both robustness and model frugality.

# Bibliographie



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